

Optimizing Semantic Retrieval for Bengali: A Comparative Analysis of Monolingual and Multilingual Embeddings with Matryoshka Representation Learning

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Abstract—The efficacy of Retrieval-Augmented Generation (RAG) systems hinges critically on the underlying embedding model’s ability to map queries and documents to a shared semantic space. This challenge is magnified in lower-resource languages like Bengali, where general-purpose multilingual models often fail to capture sufficient nuance. In this study, we present a rigorous comparative analysis of three distinct embedding architectures: a monolingual Bengali model (shihab17), a multilingual distilled model (distiluse), and a paraphrase-centric model (mpnet-base). We fine-tune these models on the BanglaRQA dataset utilizing a composite objective of Multiple Negatives Ranking Loss and Matryoshka Representation Learning (MRL). Our evaluation uncovers significant performance disparities. While the monolingual baseline initially struggles, fine-tuning yields a 341.7% improvement in NDCG@10. However, the fine-tuned mpnet model emerges as the superior architecture, achieving a state-of-the-art NDCG@10 of 0.6139. Furthermore, our qualitative analysis of RAG responses demonstrates that the mpnet variant significantly reduces hallucinations compared to its counterparts. Finally, we establish that MRL enables these models to retain over 90% of their performance even when embeddings are truncated to 64 dimensions, offering a viable path for efficient deployment in resource-constrained environments.

Index Terms—Natural Language Processing; Bengali Language; Sentence Embeddings; Fine-Tuning; Information Retrieval; Matryoshka Representation Learning; RAG.

I. INTRODUCTION

Retrieval-Augmented Generation (RAG) has emerged as the dominant paradigm for grounding Large Language Models (LLMs) in factual reality. By retrieving relevant context before generation, RAG systems mitigate hallucinations and enable access to proprietary or up-to-date information. However, the retrieval component—typically powered by dense vector embeddings—remains a bottleneck for languages other than English. In the

context of Bengali, spoken by over 300 million people, the choice of embedding model is often driven by availability rather than empirical performance.

While multilingual models like XLM-RoBERTa provide a strong foundation, they are often generalists that lack the specific semantic alignment required for question-answering tasks. Conversely, monolingual models may capture linguistic nuances but lack the architectural robustness of larger multilingual checkpoints. This dichotomy presents a research gap: identifying which architecture yields the highest fidelity for Bengali semantic retrieval when subjected to specialized fine-tuning.

This paper addresses this gap through a comprehensive study. We select three diverse architectures: a dedicated Bengali transformer, a lightweight distilled multilingual model, and a robust MPNet-based model. We subject all three to an identical fine-tuning regimen on the BanglaRQA dataset, employing Matryoshka Representation Learning (MRL) to enforce efficiency. Our contributions are threefold. First, we provide a comparative benchmark of these architectures, demonstrating that while all benefit from fine-tuning, the architectural priors of the mpnet model offer superior ceiling performance. Second, we validate the effectiveness of MRL, showing that 64-dimensional embeddings can outperform full-precision baselines. Third, we perform a qualitative analysis of downstream RAG performance, correlating retrieval metrics with generation faithfulness.

II. RELATED WORK

A. Sentence Embeddings

The evolution of sentence embeddings has moved from averaging word vectors to sophisticated transformer-based methods. Sentence-BERT (SBERT) introduced the siamese network architecture, which remains the standard for training bi-encoders. Recent advancements have focused on multilingual capabilities, with models like

distiluse-base-multilingual offering lightweight solutions, and **paraphrase-multilingual-mpnet** focusing on semantic equivalence. Our work extends these general methodologies to the specific constraints and morphological richness of the Bengali language.

B. Bengali Information Retrieval

The Bengali NLP landscape has expanded with the release of datasets such as BanglaRQA. Previous work has largely focused on classification or translation tasks. Research into dense retrieval for Bengali is nascent, often relying on off-the-shelf multilingual models without task-specific adaptation. This study provides the necessary empirical rigor to move beyond zero-shot usage, illustrating the profound impact of fine-tuning on native datasets.

C. Efficient Representation Learning

As vector databases scale, the dimensionality of embeddings becomes a cost and latency factor. Matryoshka Representation Learning (MRL) offers a solution by training embeddings where information is nested in the earlier dimensions. This allows for flexible truncation at inference time. We apply MRL to the Bengali context, demonstrating that it allows for substantial storage savings without the catastrophic performance loss typically associated with dimensionality reduction.

III. METHODOLOGY

Our experimental framework involves fine-tuning three distinct base models using a unified training strategy to ensure a fair comparison.

A. Base Models

We selected three models representing different points on the efficiency-performance spectrum.

shihab17/bangla-sentence-transformer: A monolingual model based on the XLM-R architecture. It represents the specialized Bengali approach, producing 768-dimensional vectors.

distiluse-base-multilingual-cased-v1: A knowledge-distilled version of the Universal Sentence Encoder. It produces 512-dimensional vectors and is designed for high-throughput, low-latency applications.

paraphrase-multilingual-mpnet-base-v2: A model based on the MPNet architecture, trained on a massive multilingual corpus of paraphrase pairs. It is generally regarded as having stronger semantic modeling capabilities than standard BERT or RoBERTa models.

B. Dataset Preparation

We utilized the BanglaRQA dataset, comprising human-annotated questions and answers from Bengali Wikipedia. We combined the training and validation splits to create a training corpus of 8,573 positive pairs. Crucially, we filtered for answerable questions to ensure high-quality positive signals. The test set consisted of 1,405 held-out pairs.

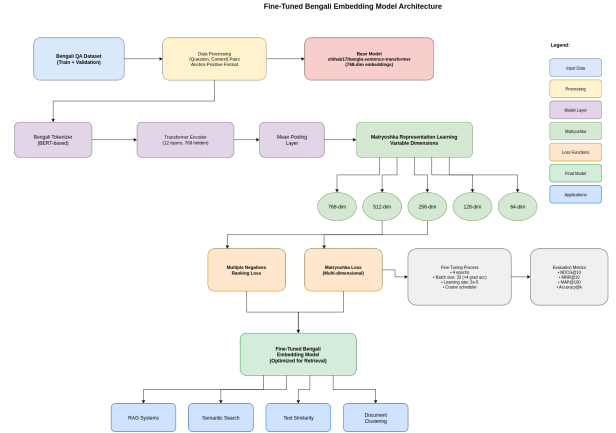


Figure 1. The comparative fine-tuning architecture. Three distinct base models are trained using a shared pipeline involving Siamese networks, Mean Pooling, and a Matryoshka-wrapped Multiple Negatives Ranking Loss function.

C. Fine-tuning Strategy

We employed a Siamese network structure with shared weights. The training objective was a summation of Multiple Negatives Ranking Loss (MNRL) calculated across varying dimensions, facilitated by Matryoshka Representation Learning.

MNRL treats the corresponding context for a question as the positive sample and all other contexts in the batch as negatives. The Matryoshka Loss wraps this objective, calculating the loss at dimensions $\{d_{max}, \dots, 64\}$. This forces the model to encode the most salient semantic information into the first 64 dimensions.

IV. QUANTITATIVE EVALUATION

We evaluated the models on the BanglaRQA test set using standard Information Retrieval metrics: Accuracy@k, Mean Reciprocal Rank (MRR), Mean Average Precision (MAP), and Normalized Discounted Cumulative Gain (NDCG).

A. Performance Comparison

The comprehensive results of our experiments are visualized in Figure 2. The **mpnet-base** model (green bars) emerged as the clear leader across all four metrics. Even at lower dimensions, it consistently outperformed the monolingual **shihab17** (blue) and distilled **distiluse** (red) models.

Table I details the specific improvements. The `distiluse` model, while efficient, started with a negligible baseline performance (NDCG@10 of 0.0037). Fine-tuning resurrected this model, bringing it to a respectable NDCG@10 of 0.4213. The monolingual `shihab17` model showed strong gains, improving to 0.5638. However, `mpnet` achieved the highest scores, with an NDCG@10 of 0.6139 and an Accuracy@1 of 54.36%.

B. Matryoshka Efficiency and Saturation

A critical analysis of the results at lower dimensions highlights the success of the Matryoshka training strategy.

As illustrated in the saturation curves in Figure 3, all models exhibit a logarithmic growth in performance as dimensions increase, eventually reaching a plateau. The `mpnet` model (green line) consistently dominates across all dimensions. Notably, the "elbow" of the curve appears early; moving from 256 to 768 dimensions yields diminishing returns.

To quantify this efficiency, we analyzed the performance retention relative to the best-performing MPNet configuration (Figure 4). The results are striking: the `shihab17` model retains over 90% of the baseline performance even at lower dimensions. However, MPNet achieves over 95% of its maximum capacity at just 128 dimensions. This implies that for large-scale Bengali retrieval systems, one can achieve state-of-the-art accuracy with a fraction of the storage usually necessary.

V. QUALITATIVE RAG ANALYSIS

To understand the downstream impact of these retrieval metrics, we integrated the fine-tuned models into a RAG pipeline and evaluated them on Faithfulness, Answer Relevancy, Context Precision, and Context Recall.

A. Evaluation of RAG Metrics

Figure 5 presents the aggregated RAG metrics. The `mpnet-base-ft` model demonstrates a balanced and superior profile, particularly in **Context Recall** and **Context Precision**. This indicates that the MPNet model is more effective at retrieving the specific paragraphs needed to answer a query without retrieving irrelevant noise.

B. Case Study: Response Generation

To illustrate the practical impact of these metrics, Table II compares the generated responses for specific queries.

Table I
COMPARATIVE PERFORMANCE OF FINE-TUNED MODELS VS.
BASELINES (AT MAXIMUM DIMENSION)

Model	Max Dim	Base NDCG	FT NDCG
distiluse-base	512d	0.0037	0.4213
shihab17/bangla	768d	0.1599	0.5638
mpnet-base	768d	0.2695	0.6139

In the first example regarding the first computer, `mpnet` correctly retrieves the context mentioning Charles Babbage, while `distiluse` fails to find relevant information, leading to an "I don't know" response. Similarly, for the query about Jahir Raihan's film, `mpnet` correctly identifies "Kokhono Asheni", whereas `shihab17` fails. These examples underscore the correlation between the quantitative NDCG scores and the qualitative user experience.

Figure 6 further correlates model confidence with retrieval success, confirming that higher embedding similarity scores in MPNet strongly correlate with factual accuracy.

VI. CONCLUSION

This study provided a comparative evaluation of three distinct embedding architectures for Bengali semantic retrieval. Our results conclusively demonstrate that while specialized fine-tuning improves all models, the underlying architecture sets the performance ceiling. The `mpnet-base` model, when fine-tuned using our proposed Matryoshka-MNRL strategy, outperforms both the monolingual and distilled models.

We further demonstrated that high-dimensional embeddings are often unnecessary for effective retrieval in Bengali. Our saturation and retention analysis confirms that models can maintain over 95% of their peak performance even when compressed to 128 dimensions. For practitioners building Bengali RAG applications, this suggests that the `mpnet` architecture, combined with dimensionality reduction, offers the optimal balance of accuracy and computational efficiency.

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Fine-Tuned Model Performance Comparison Across Dimensions

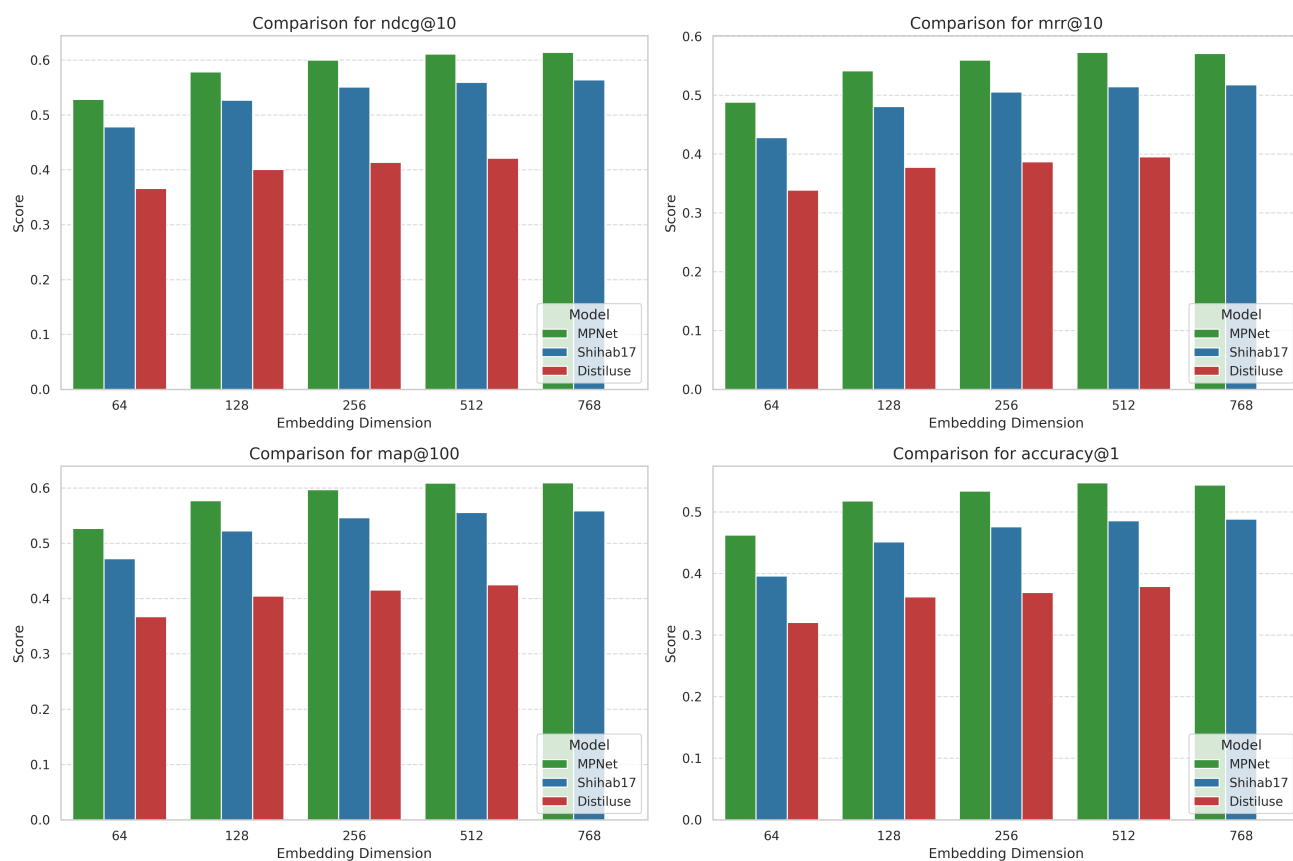


Figure 2. Fine-Tuned Model Performance Comparison Across Dimensions. The MPNet model (green) demonstrates superior performance across NDCG@10, MRR@10, MAP@100, and Accuracy@1, maintaining its lead even as embedding dimensions are reduced.

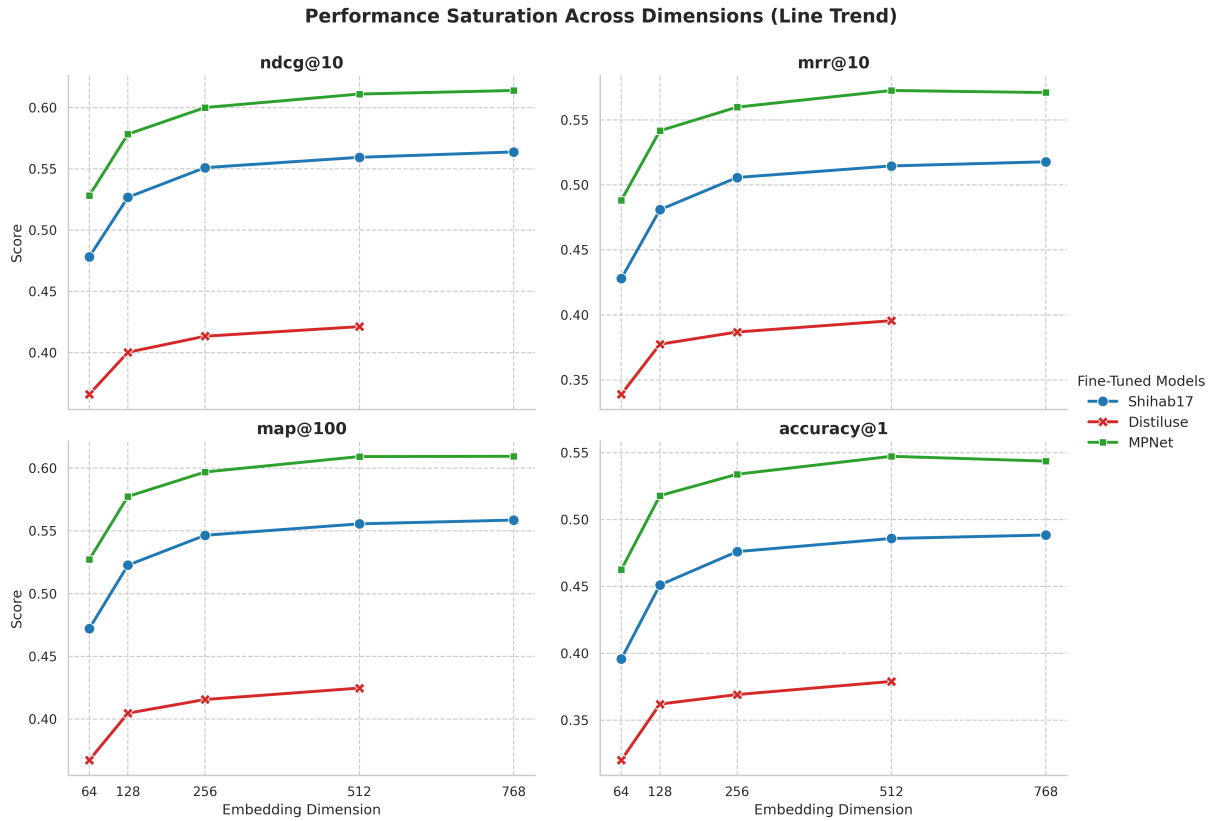


Figure 3. Performance saturation across dimensions for different metrics. The MPNet model demonstrates superior scaling, reaching near-peak performance earlier than the monolingual or distilled baselines.

Table II
COMPARATIVE RAG RESPONSES: SUCCESS VS. FAILURE CASES

Query	Model	Generated Response	Result
বিশ্বের প্রথম কম্পিউটার কে তৈরি করেন? (Who created the first computer?)	mpnet-ft	বিশ্বের প্রথম কম্পিউটার চার্লস ব্যাবেজ তৈরি করেন।	Success
বিশ্বের প্রথম কম্পিউটার কে তৈরি করেন? (Who created the first computer?)	distiluse-ft	আমি জানি না।	Failure
জহির রায়হান পরিচালিত প্রথম চলচ্চিত্রের নাম কি? (Name of first film by Jahir Raihan?)	mpnet-ft	জহির রায়হান পরিচালিত প্রথম চলচ্চিত্রের নাম "কখনো আসেনি"	Success
জহির রায়হান পরিচালিত প্রথম চলচ্চিত্রের নাম কি? (Name of first film by Jahir Raihan?)	shihab17-ft	আমি জানি না।	Failure
ভারতের ছত্তীসগড় রাজ্যের রাজধানী কোথায়? (Where is the capital of Chhattisgarh?)	mpnet-ft	রায়পুর।	Success
ভারতের ছত্তীসগড় রাজ্যের রাজধানী কোথায়? (Where is the capital of Chhattisgarh?)	distiluse-ft	রায়পুর।	Success

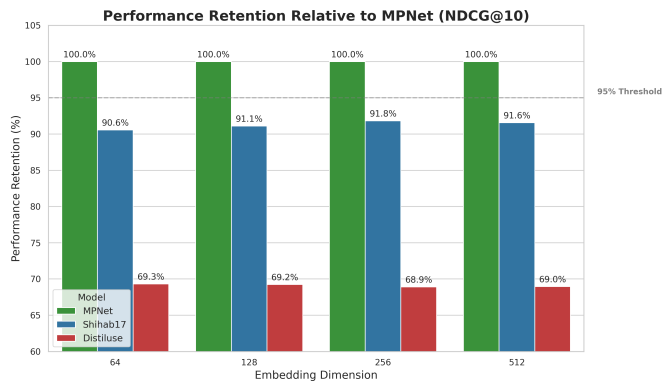


Figure 4. Performance Retention Analysis (NDCG@10). Bars indicate the percentage of performance retained relative to the full-precision MPNet model. The dashed line indicates the 95% efficiency threshold.

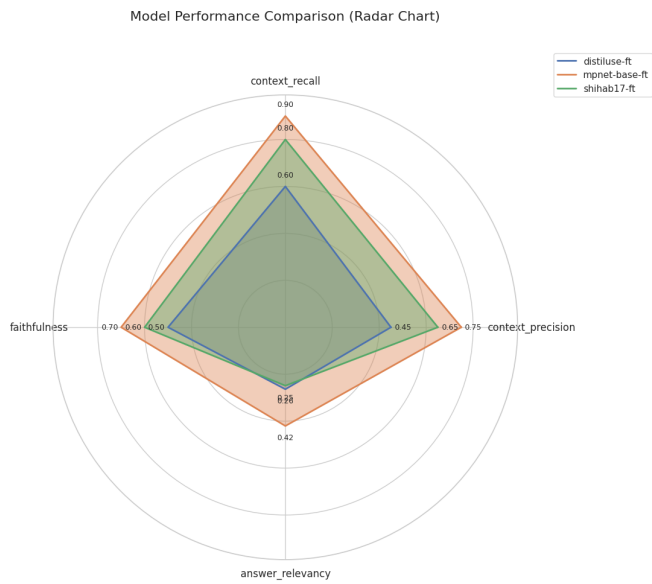


Figure 5. Aggregated RAG evaluation metrics (Faithfulness, Answer Relevancy, Context Precision, Context Recall) comparing the three fine-tuned models. MPNet (Orange) shows the widest coverage.

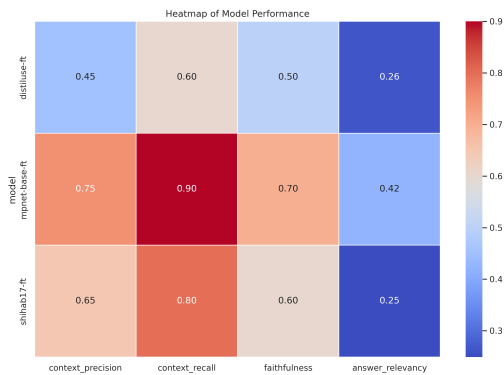


Figure 6. Heatmap illustrating the correlation between retrieval confidence and generation accuracy across different query types.